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Characteristic Functions and Moment Generating Functions

Question

Explain the difference between the characteristic function and the moment generating function.

Introduction to Generating Functions

Characteristic functions and moment generating functions are two fundamental mathematical tools in probability theory and statistics. They allow us to **completely characterize a probability distribution** and compute its moments. Despite their similarities, they present crucial differences.

Fundamental Definitions

Moment Generating Function (MGF)

The **moment generating function** of a random variable X is defined as the mathematical expectation of e^{tX}:

$$M_X(t) = \mathbb{E}[e^{tX}] = \begin{cases} \sum_x e^{tx} p_X(x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & \text{(continuous case)} \end{cases}$$

where t is a real number.

Characteristic Function

The **characteristic function** of a random variable X is defined as the mathematical expectation of e^{itX}:

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \begin{cases} \sum_x e^{itx} p_X(x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} e^{itx} f_X(x) dx & \text{(continuous case)} \end{cases}$$

where $i = \sqrt{-1}$ is the imaginary unit and t is a real number.

Detailed Comparison

1. Domain of Definition

Function	Domain of Definition	Existence
Moment Generating Function	$t \in \mathbb{R}$	**Not always** exists. Only exists if the expectation $\mathbb{E}[e^{tX}]$ converges in a neighborhood of $t = 0$
Characteristic Function	$t \in \mathbb{R}$	**Always** exists for any probability distribution

Explanation: The characteristic function uses the complex exponential e^{itx} which is always bounded ($|e^{itx}| = 1$), thus guaranteeing convergence of the integral. In contrast, e^{tx} can grow exponentially, which may prevent convergence.

2. Mathematical Nature

- **MGF**: Function with **real values** (or extended real values)
- **Characteristic Function**: Function with **complex values**

The characteristic function can be written in polar form:

$$\phi_X(t) = |\phi_X(t)| \cdot e^{i\theta(t)}$$

where $|\phi_X(t)|$ is the modulus and $\theta(t)$ is the argument.

3. Relation with Moments

For MGF

If the MGF exists in a neighborhood of $t = 0$, then moments can be obtained by differentiation:

$$\mathbb{E}[X^n] = \frac{d^n}{dt^n} M_X(t)\bigg|_{t=0}$$

For the characteristic function

Moments can also be obtained from the characteristic function (if they exist):

$$\mathbb{E}[X^n] = \frac{1}{i^n} \frac{d^n}{dt^n} \phi_X(t)\bigg|_{t=0}$$

where i^n accounts for the imaginary part.

4. Transformation Properties

Both functions have similar properties regarding linear transformations:

For $Y = aX + b$:

- **MGF**: $M_Y(t) = e^{bt} M_X(at)$
- **Characteristic Function**: $\phi_Y(t) = e^{ibt} \phi_X(at)$

5. Inversion and Uniqueness

Both functions **uniquely** characterize the distribution:

- If two random variables have the same MGF in a neighborhood of $t = 0$, then they have the same distribution
- If two random variables have the same characteristic function for all t, then they have the same distribution

However, the **inversion formula** is simpler for the characteristic function:

For a continuous distribution:

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt$$

There is no equally simple inversion formula for the MGF.

Summary Comparative Table

Aspect	Moment Generating Function (MGF)	Characteristic Function
Definition	$\mathbb{E}[e^{tX}]$	$\mathbb{E}[e^{itX}]$
Domain	Real $t \in \mathbb{R}$	Real $t \in \mathbb{R}$
Existence	Conditional (requires convergence)	**Always** guaranteed
Values	Real	Complex
Moment Calculation	$\mathbb{E}[X^n] = M_X^{(n)}(0)$	$\mathbb{E}[X^n] = \frac{\phi_X^{(n)}(0)}{i^n}$
Inversion Formula	Complex (inverse Laplace transform)	Simple (inverse Fourier transform)

Aspect	Moment Generating Function (MGF)	Characteristic Function
Main Applications	Moment calculation, theoretical proofs	Limit theorems, stochastic processes

Illustrative Examples

Example 1: Normal Distribution

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

MGF:

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \quad \text{for all } t \in \mathbb{R}$$

Characteristic Function:

$$\phi_X(t) = \exp\left(i\mu t - \frac{\sigma^2 t^2}{2}\right)$$

Note the similarity: the characteristic function is essentially the MGF with t replaced by it.

Example 2: Cauchy Distribution

The Cauchy distribution is interesting because it **has no MGF** but has a well-defined characteristic function.

For the standard Cauchy distribution:

MGF: Does not exist for $t \neq 0$ (because $\mathbb{E}[e^{tX}]$ diverges)

Characteristic Function:

$$\phi_X(t) = e^{-|t|}$$

This shows the crucial advantage of the characteristic function: it exists even for distributions without moments (or without moments of all orders).

Example 3: Exponential Distribution

For $X \sim \text{Exp}(\lambda)$ with density $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$:

MGF:

$$M_X(t) = \frac{\lambda}{\lambda - t} \quad \text{for } t < \lambda$$

Characteristic Function:

$$\phi_X(t) = \frac{\lambda}{\lambda - it}$$

Practical Applications

1. Limit Theorems

The **characteristic function** is essential for proving the central limit theorem. The standard proof uses:

1. The characteristic function of the normalized sum
2. The Taylor expansion of $\phi_X(t)$
3. Convergence to the characteristic function of the normal law

The MGF is less used because it does not always exist for normalized sums.

2. Sum of Independent Random Variables

For independent variables X and Y:

- **MGF**: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$
- **Characteristic Function**: $\phi_{X+Y}(t) = \phi_X(t) \cdot \phi_Y(t)$

This multiplicative property simplifies the study of sums.

3. Distribution Identification

If we recognize the form of an MGF or characteristic function, we can identify the corresponding distribution.

Example: If $\phi_X(t) = (1 - it/\lambda)^{-\alpha}$, then X follows a Gamma distribution with parameters α and λ.

Advantages and Disadvantages

Advantages of MGF

1. **More intuitive** for moment calculations
2. **Easier to manipulate** for distributions having moments of all orders
3. **Direct relation** with moments without imaginary factors

Disadvantages of MGF

1. **Does not always exist** (convergence problem)
2. **Limited domain** even when it exists
3. **Complex inversion formula**

Advantages of Characteristic Function

1. **Always exists** for any distribution
2. **Links with Fourier analysis** (powerful tools)
3. **Simple inversion formula**
4. **Ideal for limit theorems**

Disadvantages of Characteristic Function

1. **Complex numbers** less intuitive
2. **Moment calculation** more cumbersome (i^n factors)

Selection Guidelines

Situation	Recommended Choice
Moment calculation for distributions with all moments	MGF (simpler)
Study of distributions without certain moments	Characteristic Function
Proofs of limit theorems	Characteristic Function
Analysis of sums of independent variables	Both work
Heavy-tailed distributions (Cauchy, Pareto)	Characteristic Function

Pedagogical Conclusion

Imagine both functions as **"identity cards"** of a probability distribution:

- The **MGF** is like an identity card with practical information (the moments), but it's not always available
- The **characteristic function** is like a universal identity card that always exists, but sometimes requires a decoder (complex numbers)

Metaphor: If the probability distribution were a musical function, the MGF would be the score written in real notes, while the characteristic function would be the same score written in complex notes, capable of representing additional nuances.

In practice, statisticians and probabilists use both tools, choosing the one most suited to the specific problem. The characteristic function is generally considered more powerful and general, while the MGF is more practical when it exists.